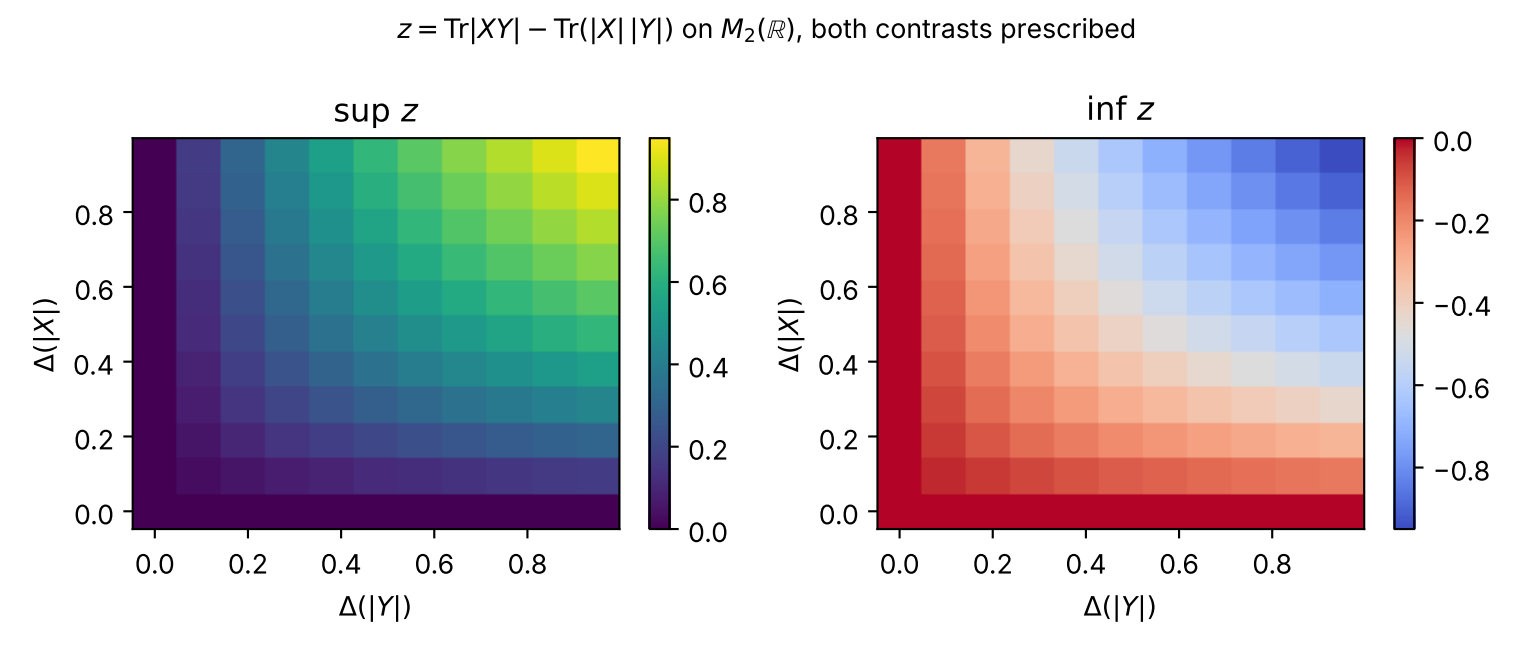
$z = \mathrm{Tr}|XY| - \mathrm{Tr}(|X|\,|Y|)$ on $M_2(\mathbb{R})$, both contrasts prescribed
sup z
inf z
$\Delta(|X|)$
$\Delta(|Y|)$
0.0
0.2
0.4
0.6
0.8
−0.2
−0.4
−0.6
−0.8